DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE; THE FACILITY OWNER'S QUALIFIED PROCEDURES AND APPLICABLE CODES GOVERN.

# WO-300-02 — Pull, clean and test the tube bundle of E-103

| | | | |
|---|---|---|---|
| **Area** | A300 Exchangers and drum | **Task** | XB Exchanger bundle pull, clean and tube-side test |
| **Priority** | 2 (integrity) | **Issued by** | Facility owner (demonstration) |
| **Primary asset** | **E-103** — E-103 shell-and-tube exchanger 36" ID × 192" — E 87'-0" N 25'-0" EL 106'-6" | | |
| **Equipment** | E-103 | **Lines** | 4"-P150-CWS-3005, 4"-P150-CWR-3006, 6"-P300-P-1011, 6"-P300-P-1012, 1"-P150-V-6027, 1"-P150-D-6028, 3/4"-P300-I-6029 |
| **Isometrics** | ISO-4"-P150-CWS-3005, ISO-4"-P150-CWR-3006, ISO-6"-P300-P-1011, ISO-6"-P300-P-1012, ISO-1"-P150-V-6027, ISO-1"-P150-D-6028, ISO-3/4"-P300-I-6029 (generated); owner's drawings ISO-CWS-3005, ISO-CWR-3006, ISO-P-1011, ISO-P-1012, ISO-V-6027, ISO-D-6028, ISO-I-6029 | | |
| **Permits** | General work permit, Confined-space entry permit, Lifting plan | | |

## Reason

Fouling: tube bundle to be pulled, cleaned and tested.

## Scope of work

1. Isolate E-103 per the isolation list (tube and shell sides); drain and vent.
2. Remove the channel cover; pull the tube bundle to the cleaning bay.
3. Clean the tubes and shell side; inspect visually; thickness at the TMLs.
4. Reinstall the bundle with new gaskets; tube-side hydrostatic test at 210 psi (SW-TEST-1).
5. Remove blinds; restore the line-up.

Affected items: E-103

Rules: SW-ITP-1, SW-ITP-2, SW-ITP-3, SW-TEST-1, SW-WO-1, SW-WO-2, SW-WO-3, SW-WO-5, SW-WO-6, SW-WO-7

## Owner's isometric drawings (SW-IMP-6)

| Drawing | Line drawn | Work line | Sheets | Match to the plant model |
|---|---|---|---|---|
| ISO-CWS-3005 | 4"-P150-CWS-3005 | 4"-P150-CWS-3005 | 1 | 0 error(s), 1 warning(s) — see below |
| ISO-CWR-3006 | 4"-P150-CWR-3006 | 4"-P150-CWR-3006 | 2 | 0 error(s), 2 warning(s) — see below |
| ISO-P-1011 | 6"-P300-P-1011 | 6"-P300-P-1011 | 2 | 0 error(s), 3 warning(s) — see below |
| ISO-P-1012 | 6"-P300-P-1012 | 6"-P300-P-1012 | 2 | 0 error(s), 2 warning(s) — see below |
| ISO-V-6027 | 1"-P150-V-6027 | 1"-P150-V-6027 | 1 | agrees with the model |
| ISO-D-6028 | 1"-P150-D-6028 | 1"-P150-D-6028 | 1 | agrees with the model |
| ISO-I-6029 | 3/4"-P300-I-6029 | 3/4"-P300-I-6029 | 1 | agrees with the model |

ISO-CWS-3005 (PDF) · ISO-CWR-3006 (PDF) · ISO-P-1011 (PDF) · ISO-P-1012 (PDF) · ISO-V-6027 (PDF) · ISO-D-6028 (PDF) · ISO-I-6029 (PDF) — the generated isometrics follow the ITP.

| Drawing | Sheet | Level | Finding |
|---|---|---|---|
| ISO-CWS-3005 | 1 | warning | weld 4"-P150-CWS-3005/W1 (E 86'-0" N 0" EL 123'-6 11/16", sheet 1) has no W1 label |
| ISO-CWR-3006 | 2 | warning | weld 4"-P150-CWR-3006/W9 (E 84'-5 5/16" N 25'-0" EL 126'-6", sheet 2) has no W9 label |
| ISO-CWR-3006 | 2 | warning | weld 4"-P150-CWR-3006/W12 (E 85'-0" N 1'-6" EL 125'-11 5/16", sheet 2) has no W12 label |
| ISO-P-1011 | 2 | warning | W14 (6"-P300-P-1011/W14, sheet 2) is labelled again on sheet 1; it is 162.0 in from the nearest break |
| ISO-P-1011 | 2 | warning | weld 6"-P300-P-1011/W20 (E 89'-0" N 25'-0" EL 110'-2 1/2", sheet 2) has no W20 label |
| ISO-P-1011 | 1 | warning | branch weld 3/4"-P300-I-7011/W1 (E 68'-11" N 6" EL 120'-0", sheet 1) has no W21 label |
| ISO-P-1012 | 2 | warning | W10 (6"-P300-P-1012/W10, sheet 2) is labelled again on sheet 1; it is 70.5 in from the nearest break |
| ISO-P-1012 | 1 | warning | branch weld 3/4"-P300-I-7012/W1 (E 99'-6" N 21'-0" EL 113'-11", sheet 1) has no W16 label |

## Assets in the plant model (SW-WO-7)

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| E-103 | equipment | E-103 shell-and-tube exchanger 36" ID × 192" | | E 87'-0" N 25'-0" EL 106'-6" |

| Tag | Kind | Description | Line | Location |
|---|---|---|---|---|
| 4"-P150-CWS-3005/B2 | item | VALVE, GATE, WEDGE, OS&Y 4" P150 | 4"-P150-CWS-3005 | E 86'-0" N 25'-0" EL 109'-3-13/16" |
| E-103-N1 | nozzle | Nozzle E-103-N1 (tube in, 4" P150) on E-103 | | E 86'-0" N 25'-0" EL 108'-8-1/16" |
| 4"-P150-CWR-3006/A2 | item | VALVE, GATE, WEDGE, OS&Y 4" P150 | 4"-P150-CWR-3006 | E 86'-0" N 25'-0" EL 103'-8-3/16" |
| E-103-N2 | nozzle | Nozzle E-103-N2 (tube out, 4" P150) on E-103 | | E 86'-0" N 25'-0" EL 104'-3-15/16" |
| 6"-P300-P-1011/B2 | item | VALVE, GATE, WEDGE, OS&Y 6" P300 | 6"-P300-P-1011 | E 89'-0" N 25'-0" EL 109'-3-7/16" |
| E-103-N3 | nozzle | Nozzle E-103-N3 (shell in, 6" P300) on E-103 | | E 89'-0" N 25'-0" EL 108'-8-5/16" |
| 6"-P300-P-1012/A2 | item | VALVE, GATE, WEDGE, OS&Y 6" P300 | 6"-P300-P-1012 | E 101'-0" N 25'-0" EL 103'-8-9/16" |
| E-103-N4 | nozzle | Nozzle E-103-N4 (shell out, 6" P300) on E-103 | | E 101'-0" N 25'-0" EL 104'-3-3/4" |
| 1"-P150-V-6027/2 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-V-6027 | E 100'-0" N 25'-0" EL 109'-1/2" |
| E-103-N5 | nozzle | Nozzle E-103-N5 (vent, 1" P150) on E-103 | | E 100'-0" N 25'-0" EL 108'-5-11/16" |
| 1"-P150-D-6028/2 | item | VALVE, BALL, FLOATING 1" P150 | 1"-P150-D-6028 | E 89'-0" N 25'-0" EL 103'-11-1/2" |
| E-103-N6 | nozzle | Nozzle E-103-N6 (drain, 1" P150) on E-103 | | E 89'-0" N 25'-0" EL 104'-6-3/8" |
| 3/4"-P300-I-6029/2 | item | VALVE, BALL, FLOATING 3/4" P300 | 3/4"-P300-I-6029 | E 95'-0" N 27'-6-1/4" EL 106'-6" |
| E-103-N7 | nozzle | Nozzle E-103-N7 (instrument, 0.75" P300) on E-103 | | E 95'-0" N 26'-11-9/16" EL 106'-6" |

## Isolation list (SW-WO-3)

| # | Kind | Tag | Line | Action | Location | Fitted ✓ | Removed ✓ |
|---|---|---|---|---|---|---|---|
| 1 | valve | 4"-P150-CWS-3005/B2 | 4"-P150-CWS-3005 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWS-3005/B2) on E-103-N1 | E 86'-0" N 25'-0" EL 109'-3-13/16" | | |
| 2 | blind | E-103-N1 | 4"-P150-CWS-3005 | Fit a blind between 4"-P150-CWS-3005/B2 and the nozzle flange E-103-N1 | E 86'-0" N 25'-0" EL 108'-8-1/16" | | |
| 3 | valve | 4"-P150-CWR-3006/A2 | 4"-P150-CWR-3006 | Close and lock VALVE, GATE, WEDGE, OS&Y 4" P150 (4"-P150-CWR-3006/A2) on E-103-N2 | E 86'-0" N 25'-0" EL 103'-8-3/16" | | |
| 4 | blind | E-103-N2 | 4"-P150-CWR-3006 | Fit a blind between 4"-P150-CWR-3006/A2 and the nozzle flange E-103-N2 | E 86'-0" N 25'-0" EL 104'-3-15/16" | | |
| 5 | valve | 6"-P300-P-1011/B2 | 6"-P300-P-1011 | Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1011/B2) on E-103-N3 | E 89'-0" N 25'-0" EL 109'-3-7/16" | | |
| 6 | blind | E-103-N3 | 6"-P300-P-1011 | Fit a blind between 6"-P300-P-1011/B2 and the nozzle flange E-103-N3 | E 89'-0" N 25'-0" EL 108'-8-5/16" | | |
| 7 | valve | 6"-P300-P-1012/A2 | 6"-P300-P-1012 | Close and lock VALVE, GATE, WEDGE, OS&Y 6" P300 (6"-P300-P-1012/A2) on E-103-N4 | E 101'-0" N 25'-0" EL 103'-8-9/16" | | |
| 8 | blind | E-103-N4 | 6"-P300-P-1012 | Fit a blind between 6"-P300-P-1012/A2 and the nozzle flange E-103-N4 | E 101'-0" N 25'-0" EL 104'-3-3/4" | | |
| 9 | valve | 1"-P150-V-6027/2 | 1"-P150-V-6027 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-V-6027/2) on E-103-N5 | E 100'-0" N 25'-0" EL 109'-1/2" | | |
| 10 | blind | E-103-N5 | 1"-P150-V-6027 | Fit a blind between 1"-P150-V-6027/2 and the nozzle flange E-103-N5 | E 100'-0" N 25'-0" EL 108'-5-11/16" | | |
| 11 | valve | 1"-P150-D-6028/2 | 1"-P150-D-6028 | Close and lock VALVE, BALL, FLOATING 1" P150 (1"-P150-D-6028/2) on E-103-N6 | E 89'-0" N 25'-0" EL 103'-11-1/2" | | |
| 12 | blind | E-103-N6 | 1"-P150-D-6028 | Fit a blind between 1"-P150-D-6028/2 and the nozzle flange E-103-N6 | E 89'-0" N 25'-0" EL 104'-6-3/8" | | |
| 13 | valve | 3/4"-P300-I-6029/2 | 3/4"-P300-I-6029 | Close and lock VALVE, BALL, FLOATING 3/4" P300 (3/4"-P300-I-6029/2) on E-103-N7 | E 95'-0" N 27'-6-1/4" EL 106'-6" | | |
| 14 | blind | E-103-N7 | 3/4"-P300-I-6029 | Fit a blind between 3/4"-P300-I-6029/2 and the nozzle flange E-103-N7 | E 95'-0" N 26'-11-9/16" EL 106'-6" | | |

## Inspections

| Id | Inspection | Criteria | Basis |
|---|---|---|---|
| WO-300-02-I1 | Internal visual examination of the shell, channel and tube bundle | Findings recorded against the owner's integrity criteria | SW-NDE-4 |
| WO-300-02-I2 | Ultrasonic thickness at the 12 TMLs listed | Each reading above the owner's minimum wall | SW-WO-6 |

## Thickness monitoring locations (SW-WO-6)

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-300-02-TML01 | E-103 | Nozzle neck E-103-N1 | E 86'-0" N 25'-0" EL 108'-8-1/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-02-TML02 | E-103 | Nozzle neck E-103-N2 | E 86'-0" N 25'-0" EL 104'-3-15/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-02-TML03 | E-103 | Nozzle neck E-103-N3 | E 89'-0" N 25'-0" EL 108'-8-5/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-02-TML04 | E-103 | Nozzle neck E-103-N4 | E 101'-0" N 25'-0" EL 104'-3-3/4" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-02-TML05 | E-103 | Nozzle neck E-103-N5 | E 100'-0" N 25'-0" EL 108'-5-11/16" | | top: ____ | side: ____ | bottom: ____ | side: ____ |

| TML | Item | Location | Position | Nominal wall | R1 | R2 | R3 | R4 |
|---|---|---|---|---|---|---|---|---|
| WO-300-02-TML06 | E-103 | Nozzle neck E-103-N6 | E 89'-0" N 25'-0" EL 104'-6-3/8" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-02-TML07 | E-103 | Nozzle neck E-103-N7 | E 95'-0" N 26'-11-9/16" EL 106'-6" | | top: ____ | side: ____ | bottom: ____ | side: ____ |
| WO-300-02-TML08 | E-103 | Left head centre | E 86'-2-3/4" N 25'-0" EL 106'-6" | | centre: ____ | | | |
| WO-300-02-TML09 | E-103 | Right head centre | E 103'-9-1/4" N 25'-0" EL 106'-6" | | centre: ____ | | | |
| WO-300-02-TML10 | E-103 | Shell at 0.000" from the left tangent | E 87'-0" N 25'-0" EL 106'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-300-02-TML11 | E-103 | Shell at 6'-0" from the left tangent | E 93'-0" N 25'-0" EL 106'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |
| WO-300-02-TML12 | E-103 | Shell at 12'-0" from the left tangent | E 99'-0" N 25'-0" EL 106'-6" | | 0°: ____ | 90°: ____ | 180°: ____ | 270°: ____ |

## Test — Hydrostatic test (tube side)

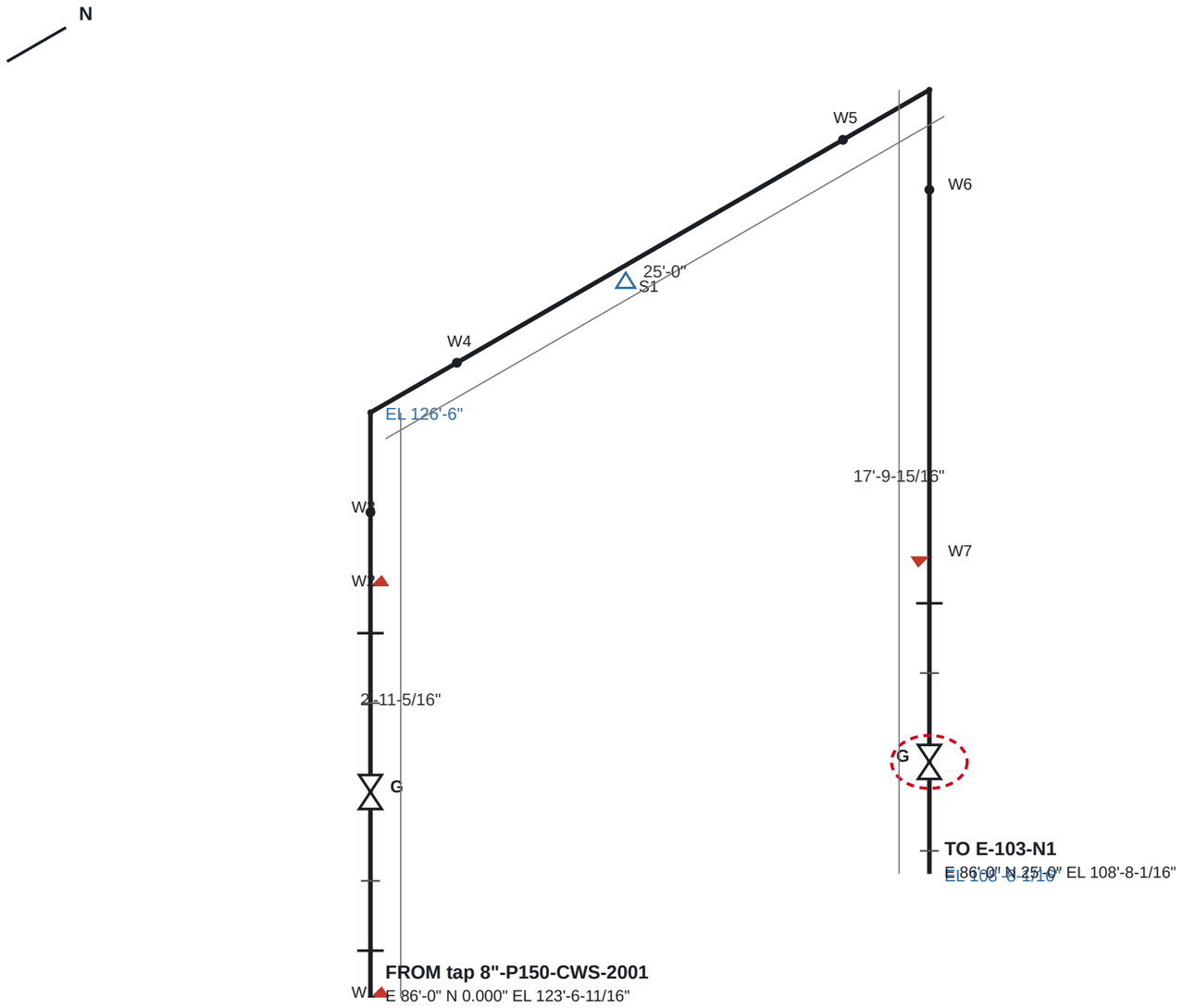
| | |
|---|---|
| **Description** | Tube side of the reinstalled bundle |
| **Pressure** | 210 psi |
| **Hold** | 30 minutes, no leak, no pressure drop |
| **Basis** | SW-TEST-1 |

## Inspection and test plan (SW-ITP-1..3)

H hold · W witness · R review · S surveillance. Status: accepted / rejected / not applicable.

| # | Activity | Reference | Acceptance | Record | Contr. | Owner | 3rd party | Status | Signed (contr. / owner / 3rd) | Date | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Work permit issued; isolation list (valves locked, blinds fitted) verified at zero pressure and drained | SW-WO-3, isolation list | Every isolation in place and tagged; zero pressure at the vent | Signed isolation certificate | H | H | — | | | | |
| 2 | Equipment vented, purged and gas-tested for entry; confined-space entry permit issued | Owner entry procedure | Atmosphere safe for entry per owner limits | Gas test record, entry permit | H | H | — | | | | |
| 3 | Channel cover removed; bundle pulled and cleaned | Owner procedure | Surfaces clean enough to inspect | Daily report | S | S | — | | | | |
| 4 | Internal visual examination | SW-NDE-4 | Findings recorded | Inspection report with photographs | W | H | W | | | | |
| 5 | Ultrasonic thickness at 12 TMLs | SW-WO-6 | Above the owner's minimum wall | Thickness record per TML | S | R | R | | | | |
| 6 | Owner's decision on findings (repair, monitor or accept) | Owner integrity procedure | Decision recorded | Signed disposition | R | H | R | | | | |
| 7 | Closed with new gaskets on every opened joint; studs to preload | SW-WO-5 | Preload as stated | Bolt-up record | W | W | — | | | | |
| 8 | Tube-side hydrostatic test at 210 psi, 30 minutes | SW-TEST-1 | No leak, no pressure drop | Test record | H | H | H | | | | |
| 9 | Blinds removed, isolations returned to normal line-up per the isolation list | SW-WO-3 | Every blind removed and accounted for | Isolation certificate closed | H | H | — | | | | |
| 10 | Records complete; work order handed back to operations | SW-ITP-3 | All ITP activities accepted or not applicable | Signed ITP and completion record | H | H | R | | | | |

ISOMETRIC — 4"-P150-CWS-3005 — Pull, clean and test the tube bundle of E-103




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | FLANGE WELD-NECK 4" P150 RAISED FACE | 3 |
| 2 | GASKET 4" P150 COMPRESSED SHEET | 4 |
| 3 | VALVE, GATE, WEDGE, OS&Y 4" P150 | 2 |
| 4 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150 | 4 |
| 5 | PIPE 4" P150 BEVELLED ENDS | 39'-7-5/16" |
| 6 | ELBOW 90° R = 5/3 OD 4" P150 | 2 |

**WELDS 7 (3 FIELD) · SUPPORTS 1 · SPOOLS 4**

W1F W2F W3 W4 W5 W6 W7F

## 4"-P150-CWS-3005

CLASS W150 — Water (cooling and process)
SERVICE CWS · 4" · P150 · DESIGN 150 psi
tap 8"-P150-CWS-2001 → E-103-N1
PIPE 39'-7-5/16" · WEIGHT 693 lb

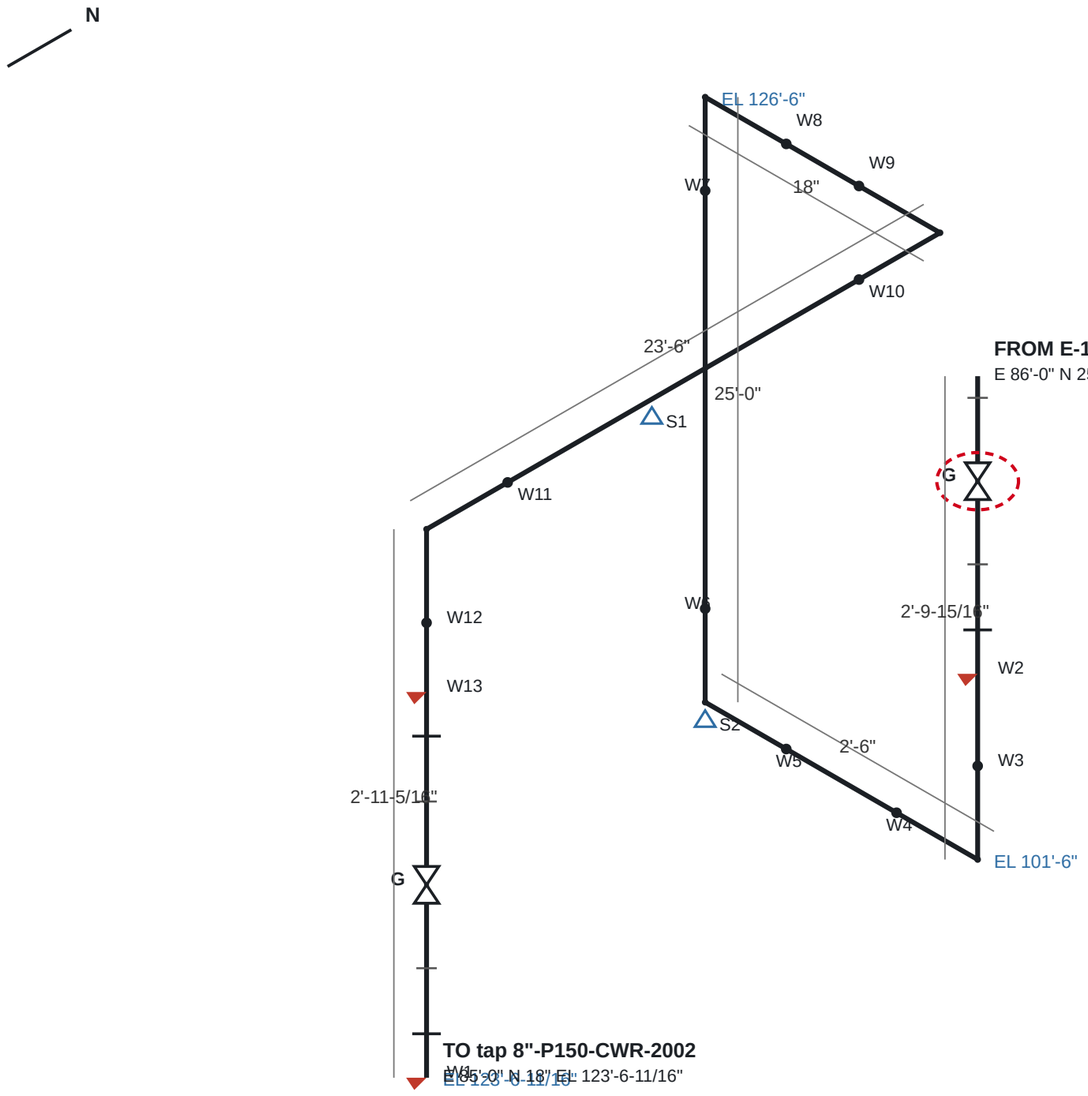
**DWG ISO-4"-P150-CWS-3005 REV 0 WO-300-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

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## Isometric ISO-4"-P150-CWR-3006

**ISOMETRIC — 4"-P150-CWR-3006 — Pull, clean and test the tube bundle of E-103**




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 4" P150 COMPRESSED SHEET | 4 |
| 2 | VALVE, GATE, WEDGE, OS&Y 4" P150 | 2 |
| 3 | STUD SET 4 × 1" × 6-1/4" WITH 2 NUTS, FOR 4" P150 | 4 |
| 4 | FLANGE WELD-NECK 4" P150 RAISED FACE | 3 |
| 5 | PIPE 4" P150 BEVELLED ENDS | 48'-9-5/16" |
| 6 | ELBOW 90° R = 5/3 OD 4" P150 | 5 |

## WELDS 13 (3 FIELD) · SUPPORTS 2 · SPOOLS 4

W1F W2F W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F

## 4"-P150-CWR-3006

CLASS W150 — Water (cooling and process)
SERVICE CWR · 4" · P150 · DESIGN 150 psi
E-103-N2 → tap 8"-P150-CWR-2002
PIPE 48'-9-5/16" · WEIGHT 783 lb

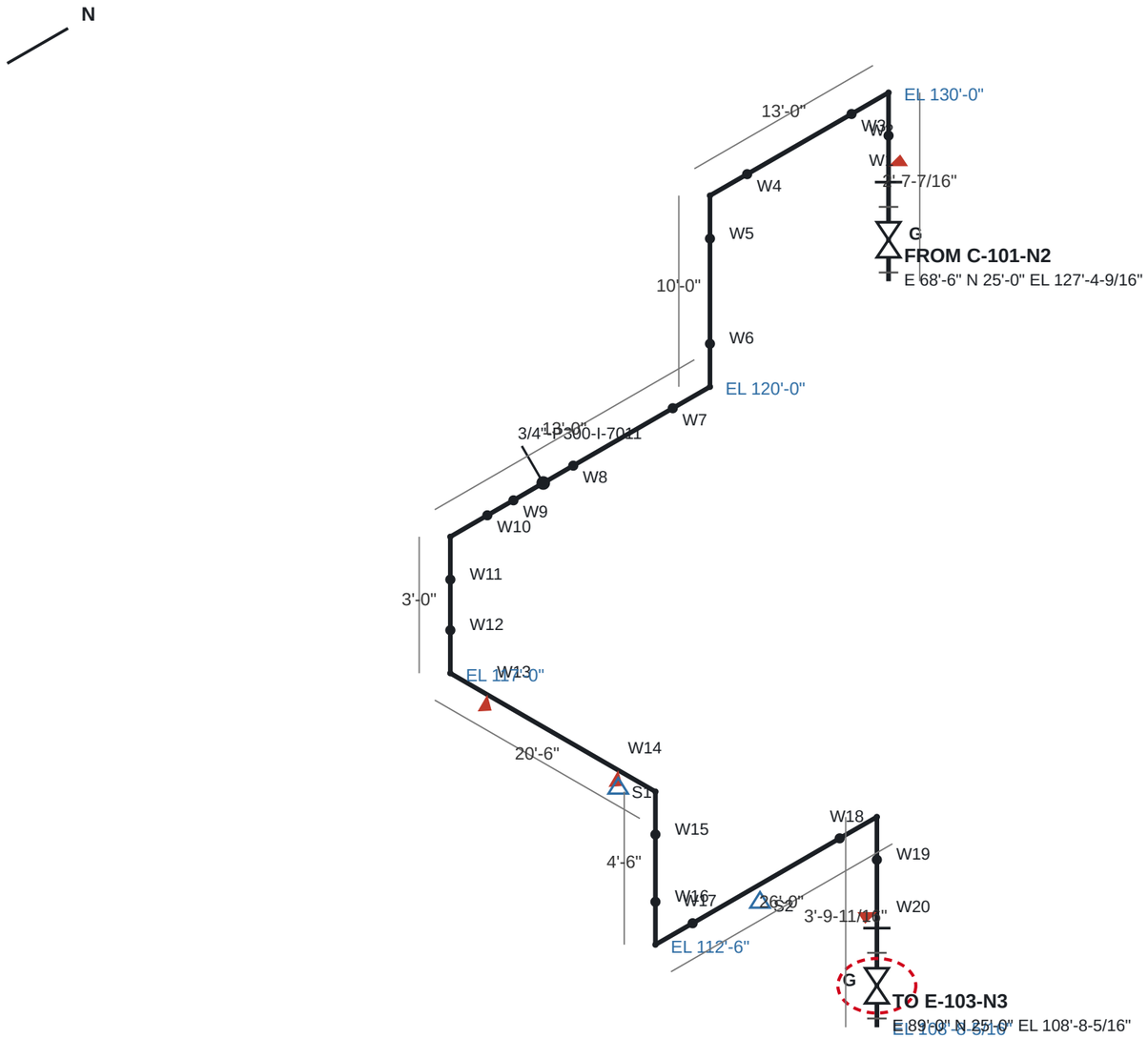
**DWG ISO-4"-P150-CWR-3006 REV 0 WO-300-02**

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## Isometric ISO-6"-P300-P-1011

## ISOMETRIC — 6"-P300-P-1011 — Pull, clean and test the tube bundle of E-103




### BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 6" P300 COMPRESSED SHEET | 4 |
| 2 | VALVE, GATE, WEDGE, OS&Y 6" P300 | 2 |
| 3 | STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300 | 4 |
| 4 | FLANGE WELD-NECK 6" P300 RAISED FACE | 2 |
| 5 | PIPE 6" P300 BEVELLED ENDS | 79'-2-13/16" |
| 6 | ELBOW 90° R = 5/3 OD 6" P300 | 8 |
| 7 | TEE REDUCING 6" × 3/4" P300 | 1 |

### WELDS 20 (4 FIELD) · SUPPORTS 2 · SPOOLS 5

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13F W14F W15 W16 W17 W18 W19 W20F

## 6"-P300-P-1011

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
C-101-N2 → E-103-N3
PIPE 79'-2-13/16" · WEIGHT 1677 lb

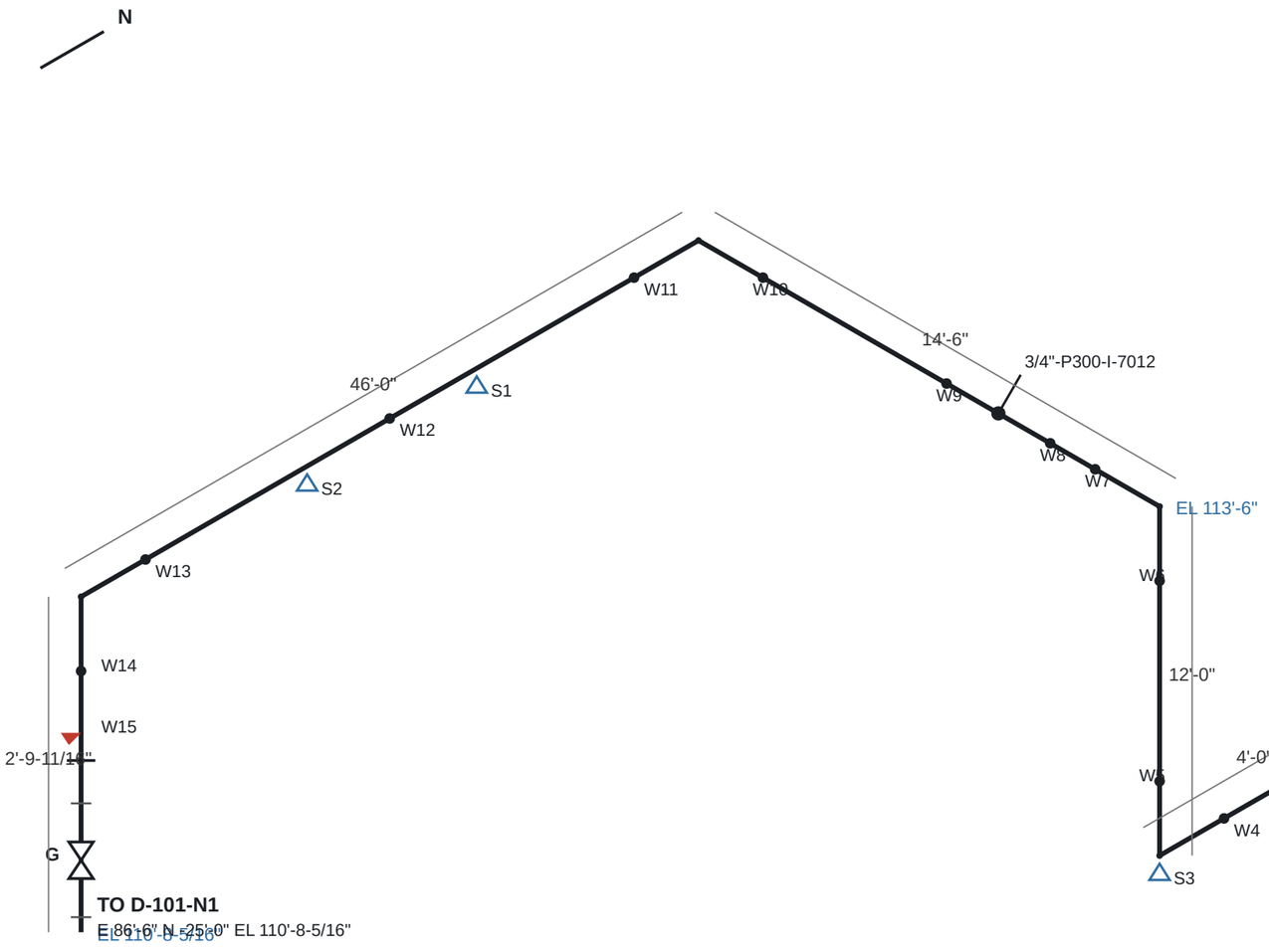
**DWG ISO-6"-P300-P-1011 REV 0 WO-300-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**

## Isometric ISO-6"-P300-P-1012

**ISOMETRIC — 6"-P300-P-1012 — Pull, clean and test the tube bundle of E-103**




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 6" P300 COMPRESSED SHEET | 4 |
| 2 | VALVE, GATE, WEDGE, OS&Y 6" P300 | 2 |
| 3 | STUD SET 8 × 0.625" × 4" WITH 2 NUTS, FOR 6" P300 | 4 |
| 4 | FLANGE WELD-NECK 6" P300 RAISED FACE | 2 |
| 5 | PIPE 6" P300 BEVELLED ENDS | 69'-11-1/16" |
| 6 | ELBOW 90° R = 5/3 OD 6" P300 | 5 |
| 7 | TEE REDUCING 6" × 3/4" P300 | 1 |

**WELDS 15 (2 FIELD) · SUPPORTS 3 · SPOOLS 3**

W1F W2 W3 W4 W5 W6 W7 W8 W9 W10
W11 W12 W13 W14 W15F

## 6"-P300-P-1012

CLASS H300 — Hydrocarbon process
SERVICE P · 6" · P300 · DESIGN 300 psi
E-103-N4 → D-101-N1
PIPE 69'-11-1/16" · WEIGHT 1498 lb

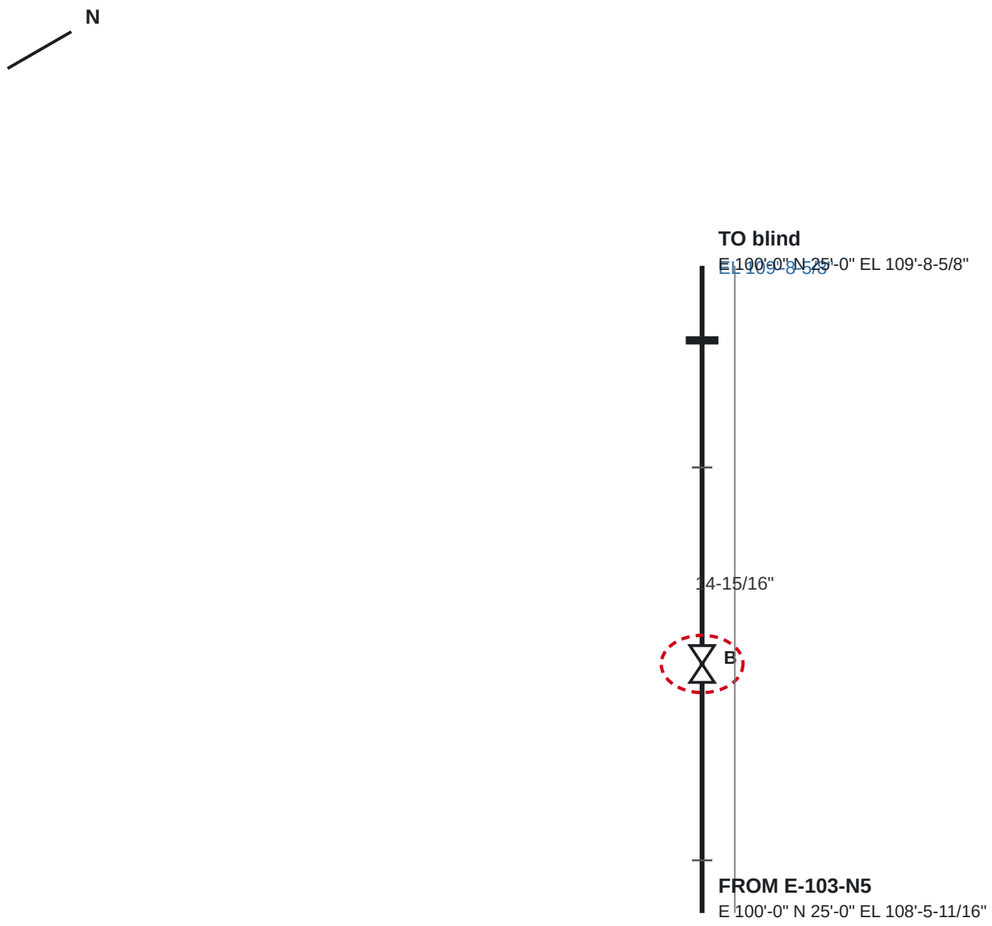
**DWG ISO-6"-P300-P-1012 REV 0 WO-300-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**

## Isometric ISO-1"-P150-V-6027

ISOMETRIC — 1"-P150-V-6027 — Pull, clean and test the tube bundle of E-103




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 1" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 1" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 2 |
| 4 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 1"-P150-V-6027

CLASS H150 — Hydrocarbon process
SERVICE V · 1" · P150 · DESIGN 150 psi
E-103-N5 → blind
PIPE 0.000" · WEIGHT 39 lb

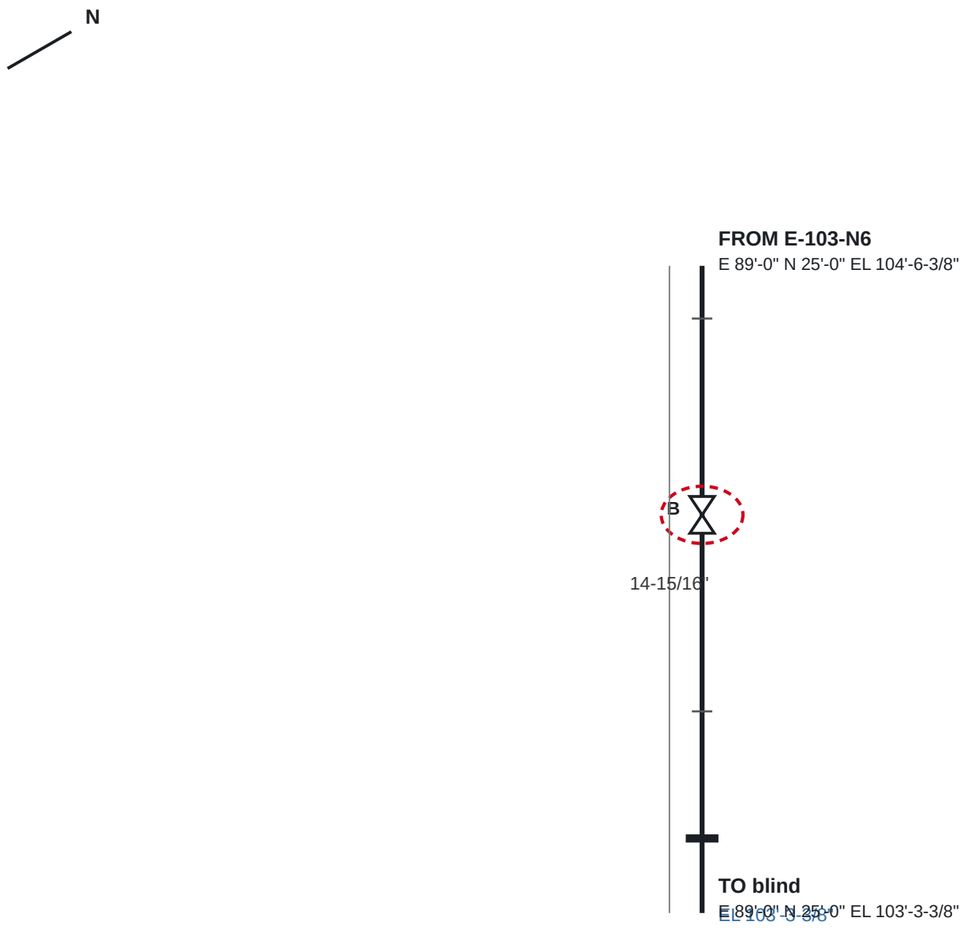
**DWG ISO-1"-P150-V-6027 REV 0 WO-300-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**

## Isometric ISO-1"-P150-D-6028

**ISOMETRIC — 1"-P150-D-6028 — Pull, clean and test the tube bundle of E-103**




**BILL OF MATERIALS**

| | | |
|---|---|---|
| 1 | GASKET 1" P150 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 1" P150 | 1 |
| 3 | STUD SET 4 × 0.5" × 4" WITH 2 NUTS, FOR 1" P150 | 2 |
| 4 | FLANGE BLIND 1" P150 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 1"-P150-D-6028

CLASS H150 — Hydrocarbon process
SERVICE D · 1" · P150 · DESIGN 150 psi
E-103-N6 → blind
PIPE 0.000" · WEIGHT 39 lb

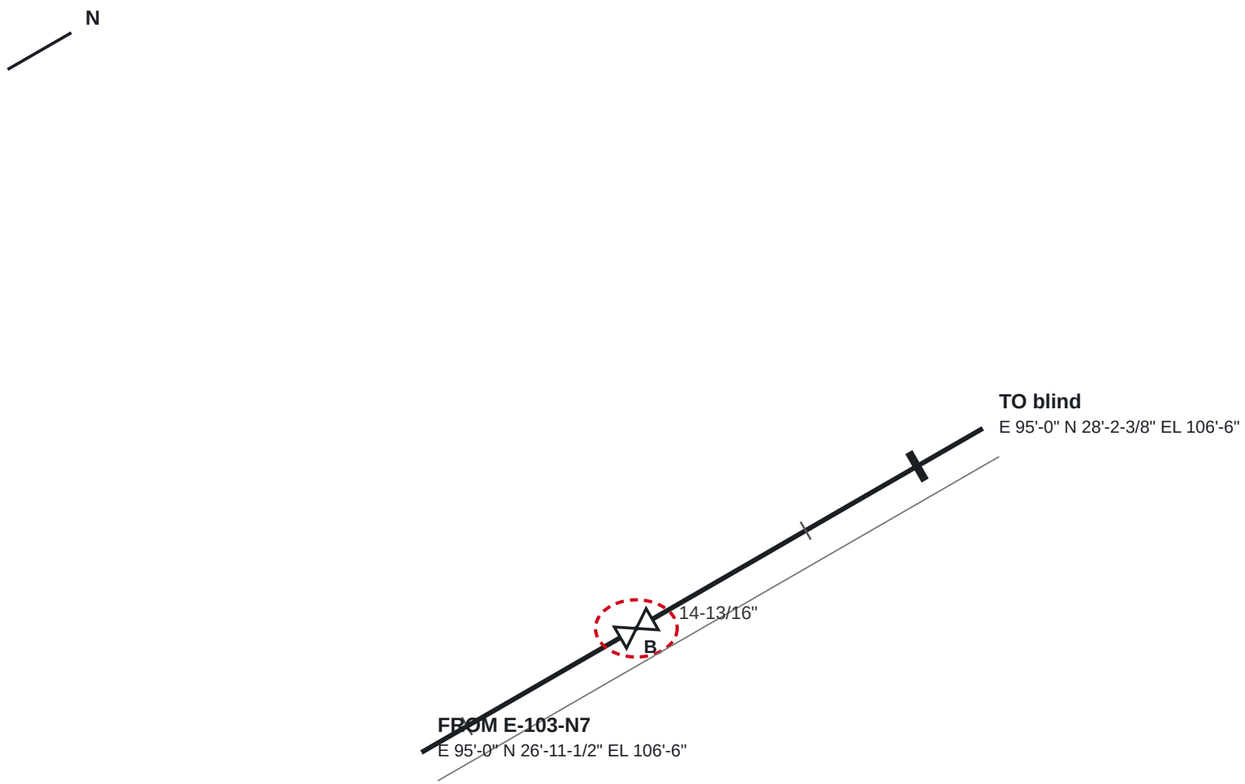
**DWG ISO-1"-P150-D-6028 REV 0 WO-300-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**

## Isometric ISO-3/4"-P300-I-6029

**ISOMETRIC — 3/4"-P300-I-6029 — Pull, clean and test the tube bundle of E-103**




## BILL OF MATERIALS

| | | |
|---|---|---|
| 1 | GASKET 3/4" P300 COMPRESSED SHEET | 2 |
| 2 | VALVE, BALL, FLOATING 3/4" P300 | 1 |
| 3 | STUD SET 4 × 0.5" × 4-1/4" WITH 2 NUTS, FOR 3/4" P300 | 2 |
| 4 | FLANGE BLIND 3/4" P300 RAISED FACE | 1 |

**WELDS 0 (0 FIELD) · SUPPORTS 0 · SPOOLS 0**

## 3/4"-P300-I-6029

CLASS H300 — Hydrocarbon process
SERVICE I · 0.75" · P300 · DESIGN 300 psi
E-103-N7 → blind
PIPE 0.000" · WEIGHT 32 lb

**DWG ISO-3/4"-P300-I-6029 REV 0 WO-300-02**

SmartWeld AI · not to scale · E lower right, N upper right, EL up (SW-ISO-1)

**DEMONSTRATION ONLY — NOT A QUALIFIED PROCEDURE**